ONCO-PRISM: PageRank-Guided Graph Attention for Cancer Region Importance in Whole Slide Images

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Abstract

Whole Slide Image (WSI) analysis in histopathology is a fundamental task in cancer diagnosis. Existing approaches either rely on convolutional or transformer-based models that treat image patches independently, or graph-based methods that fail to incorporate global structural importance. In this work, we propose **ONCO-PRISM**, a novel framework that models tissue as a graph and integrates *PageRank-driven attention* within Graph Neural Networks (GNNs). Our approach captures both local cellular interactions and global structural importance, enabling improved identification of diagnostically relevant regions. We demonstrate that incorporating PageRank as a structural prior enhances interpretability and improves performance over standard Vision Transformers and Graph Attention Networks. Extensive experiments on histopathology datasets validate the effectiveness of the proposed approach.

1 Introduction

Histopathology remains the gold standard for cancer diagnosis, providing detailed visual evidence of tissue morphology and cellular organization. With the digitization of pathology workflows, Whole Slide Images (WSIs) have emerged as a rich source of data, often reaching gigapixel resolutions. While these images contain critical diagnostic information, their sheer size and complexity pose significant computational and analytical challenges. Consequently, most modern approaches adopt a patch-based paradigm, where WSIs are divided into smaller regions that can be processed using deep learning models.

Early methods predominantly relied on convolutional neural networks (CNNs) to classify individual patches, followed by aggregation schemes to obtain slide-level predictions. More recently, Vision Transformers (ViTs) have demonstrated improved capability in capturing long-range dependencies within patches. However, despite these advances, both CNN- and transformer-based approaches fundamentally treat patches as independent or weakly related entities. This assumption neglects the inherent spatial and structural relationships between different tissue regions, which are often crucial for accurate diagnosis. In particular, cancer progression is not merely a local phenomenon but is reflected in complex spatial interactions across tissue microenvironments.

To address this limitation, recent works have explored graph-based representations of histopathology images, where patches are treated as nodes and edges encode spatial or feature-based relationships. Graph Neural Networks (GNNs) operating on such structures enable relational reasoning and have shown promise in modeling tissue organization. Nevertheless, existing GNN-based approaches typically assume uniform importance across nodes or rely solely on learned attention mechanisms. As a result, they lack an explicit notion of global structural importance, which can guide the model toward diagnostically significant regions.

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In parallel, graph theory offers well-established measures of node importance, among which PageRank has been widely used to quantify the influence of nodes within a network. PageRank captures both local connectivity and global structural relevance, making it a natural candidate for identifying critical regions in tissue graphs. Despite its effectiveness in other domains, its integration into deep learning frameworks for histopathology analysis remains largely unexplored.

In this work, we propose ONCO-PRISM, a novel framework that integrates PageRank-based structural importance into graph neural networks for cancer region identification in WSIs. We model tissue as a graph constructed from image patches and compute PageRank scores to estimate the global importance of each region. These scores are then incorporated into a Graph Attention Network, effectively biasing the attention mechanism toward structurally significant nodes. This fusion of classical graph centrality with modern deep learning enables the model to capture both relational dependencies and global importance patterns.

Our approach offers several key advantages. First, it introduces a principled mechanism to incorporate global structural priors into GNNs, addressing a critical gap in existing methods. Second, it enhances interpretability by producing importance maps that align with clinically relevant regions. Third, it provides a unified framework that bridges computer vision, graph learning, and network science for histopathology analysis.

Through experiments on whole slide image datasets, we demonstrate that the proposed method improves both classification performance and localization quality compared to standard Vision Transformer and graph-based baselines. These results highlight the potential of PageRank-guided attention as a powerful tool for medical image analysis and open new directions for integrating classical graph algorithms with deep learning models.

2 Related Work

2.1 Deep Learning for Histopathology Analysis

Deep learning has significantly advanced computational pathology, particularly through convolutional neural networks (CNNs) applied to histopathology images. Early works leveraged architectures such as ResNet [3] and DenseNet [4] for patch-level classification, followed by heuristic aggregation to obtain slide-level predictions. While effective, these approaches treat image patches independently and fail to capture contextual dependencies across tissue regions.

To address the limitations of CNNs in modeling long-range interactions, Vision Transformers (ViTs) [1] have been introduced for medical imaging tasks. By leveraging self-attention, ViTs enable global receptive fields and improved contextual reasoning. However, despite their ability to model intra-patch relationships, they do not explicitly encode spatial or structural relationships between patches extracted from whole slide images.

2.2 Multiple Instance Learning in Whole Slide Images

Multiple Instance Learning (MIL) has emerged as a dominant paradigm for weakly supervised learning in WSIs, where only slide-level labels are available. In MIL frameworks, a WSI is treated as a bag of instances (patches), and predictions are aggregated using pooling strategies. Attention-based MIL methods [5] further improve performance by assigning learned importance weights to patches. More recent works such as CLAM [7] introduce clustering constraints to enhance discriminative learning.

Despite these improvements, MIL-based approaches suffer from two major limitations. First, they do not explicitly model spatial relationships between patches. Second, the learned attention weights are purely data-driven and lack structural grounding, which limits interpretability and robustness.

2.3 Graph-Based Modeling of Histopathology Images

To incorporate spatial structure, several recent works model WSIs as graphs, where nodes correspond to patches and edges represent spatial or feature-based relationships. Graph Neural Networks (GNNs) [6] and GraphSAGE [2] have been applied to propagate contextual information across

tissue regions. These approaches enable relational reasoning and better capture tissue organization compared to independent patch-based models.

Graph Attention Networks (GATs) [9] further extend GNNs by introducing learnable attention mechanisms over graph edges. While GATs allow adaptive weighting of neighboring nodes, the attention mechanism is entirely learned from data and does not incorporate any explicit notion of global importance. Consequently, these models may fail to identify structurally critical regions in complex tissue graphs.

2.4 Graph Centrality and Structural Importance

Graph centrality measures, such as PageRank [8], provide a principled way to quantify the global importance of nodes within a network. PageRank has been widely used in web search, social network analysis, and biological networks to identify influential nodes. Its formulation captures both local connectivity and global structural influence through recursive propagation.

Despite its success in classical graph analysis, the integration of PageRank into deep learning frameworks remains relatively underexplored. A few works have investigated incorporating centrality measures into graph learning, but their application to medical imaging and histopathology is still limited.

2.5 Limitations of Existing Approaches

From the above discussion, we identify three key limitations in existing methods:

- **Lack of structural modeling:** CNNs, ViTs, and MIL approaches do not explicitly capture spatial relationships between tissue regions.
- **Absence of global importance:** Existing GNN and GAT models rely solely on learned attention without incorporating global structural priors.
- **Limited interpretability:** Attention mechanisms in current models are difficult to interpret without a principled notion of importance.

2.6 Positioning of Our Work

To address these limitations, we propose integrating PageRank-based structural importance into graph neural networks for histopathology analysis. By combining classical graph centrality with modern deep learning, our approach introduces a principled and interpretable mechanism for identifying diagnostically relevant regions in WSIs. This enables the model to leverage both relational context and global structural information, bridging a critical gap in existing literature.

3 Problem Formulation

Whole Slide Images (WSIs) are high-resolution histopathology images that can be represented as continuous pixel-valued functions. Formally, a WSI is defined as:

$$I \in \mathbb{R}^{H \times W \times 3} \tag{1}$$

where H and W denote the spatial dimensions and the three channels correspond to RGB intensities. Due to the extremely large size of WSIs, direct processing is computationally infeasible. Therefore, the image is partitioned into a set of smaller, fixed-size patches:

$$\mathcal{P} = \{p_1, p_2, \ldots, p_N\}, \quad p_i \in \mathbb{R}^{s \times s \times 3} \tag{2}$$

where N denotes the number of extracted patches and s is the patch size.

Each patch is mapped to a feature representation using a deep feature extractor $\phi(\cdot)$, typically a Vision Transformer:

$$x_i = \phi(p_i), \quad x_i \in \mathbb{R}^D \tag{3}$$

Stacking all patch embeddings yields a feature matrix:

$$X = [x_1, x_2, \ldots, x_N]^T \in \mathbb{R}^{N \times D} \tag{4}$$

To model spatial relationships between patches, we construct a graph:

$$G = (V, E) \tag{5}$$

where each node $v_i \in V$ corresponds to a patch embedding x_i, and edges $(i, j) \in E$ represent spatial or feature-based proximity between patches. The graph structure is encoded using an adjacency matrix:

$$A \in \mathbb{R}^{N \times N} \tag{6}$$

with $A_{ij} > 0$ if nodes i and j are connected.

Given this graph representation, the objective is to learn a function:

$$f : (G, X) \to y \tag{7}$$

where $y \in \{0, 1\}$ denotes the slide-level cancer label.

In addition to classification, we aim to estimate the importance of each patch in contributing to the final prediction. Let π_i denote the importance score of node i. The goal is to learn a mapping:

$$\pi_i = g(G, X, i) \tag{8}$$

such that π_i reflects the structural and contextual relevance of patch p_i.

Unlike existing approaches that rely solely on learned attention mechanisms, we introduce a structural prior over node importance using graph-based centrality. Specifically, we leverage PageRank to define a global importance distribution over nodes:

$$\boldsymbol{\pi} \in \mathbb{R}^N, \quad \sum_{i=1}^{N} \pi_i = 1 \tag{9}$$

This formulation enables the integration of global structural information into the learning process, guiding the model toward diagnostically relevant regions.

Thus, the problem can be summarized as jointly learning:

1. A graph-based representation of tissue structure,
2. A node importance function incorporating structural priors,
3. A slide-level classifier leveraging both local features and global importance.

4 Proposed Method

In this section, we present ONCO-PRISM, a framework that integrates graph-based structural modeling with PageRank-guided attention for cancer region identification in Whole Slide Images (WSIs). Our approach combines local feature learning with global structural importance to produce both accurate predictions and interpretable importance maps.

4.1 Overview

Given a WSI, our pipeline consists of the following stages:

1. Patch extraction from the WSI
2. Feature extraction using a Vision Transformer
3. Graph construction based on spatial relationships
4. Computation of PageRank-based node importance
5. Graph neural network with PageRank-guided attention
6. Slide-level prediction via aggregation

[Insert Architecture Diagram Here********

4.2 Patch Extraction and Feature Representation

The input WSI $I \in \mathbb{R}^{H \times W \times 3}$ is partitioned into N patches:

$$\mathcal{P} = \{p_1, p_2, \ldots, p_N\}, \quad p_i \in \mathbb{R}^{s \times s \times 3} \tag{10}$$

To remove background regions, patches with high mean intensity are discarded:

$$\frac{1}{s^2} \sum_{u,v} p_i(u,v) > \tau \Rightarrow p_i \text{ is removed} \tag{11}$$

Each patch is then mapped to a feature vector using a Vision Transformer $\phi(\cdot)$:

$$x_i = \phi(p_i), \quad x_i \in \mathbb{R}^D \tag{12}$$

The resulting feature matrix is:

$$X \in \mathbb{R}^{N \times D} \tag{13}$$

4.3 Graph Construction

To model spatial relationships between patches, we construct a graph $G = (V, E)$ where each node corresponds to a patch embedding. Edges are defined using a k-nearest neighbor (kNN) graph:

$$(i,j) \in E \iff j \in \mathcal{N}_k(i) \tag{14}$$

The adjacency matrix $A \in \mathbb{R}^{N \times N}$ encodes the connectivity structure of the graph.

4.4 PageRank-Based Structural Importance

We compute a global importance score for each node using PageRank. Let π_i denote the importance of node i. PageRank is defined recursively as:

$$\pi_i = \frac{1-\alpha}{N} + \alpha \sum_{j \in \mathcal{N}(i)} \frac{\pi_j}{\deg(j)} \tag{15}$$

where $\alpha \in (0,1)$ is the damping factor and $\deg(j)$ is the degree of node j.

In matrix form:

$$\boldsymbol{\pi} = \alpha P^T \boldsymbol{\pi} + (1-\alpha)\frac{1}{N}\mathbf{1} \tag{16}$$

where P is the transition probability matrix derived from A.

The resulting vector $\boldsymbol{\pi} \in \mathbb{R}^N$ provides a normalized measure of global structural importance across the graph.

4.5 PageRank-Guided Graph Attention

We integrate PageRank into a Graph Attention Network (GAT) to guide the learning process. In standard GAT, attention coefficients are computed as:

$$e_{ij} = \text{LeakyReLU}\left(\mathbf{a}^T[Wx_i \,\|\, Wx_j]\right) \tag{17}$$

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k\in\mathcal{N}(i)} \exp(e_{ik})} \tag{18}$$

To incorporate structural importance, we modulate the attention coefficients using PageRank:

$$\tilde{e}_{ij} = e_{ij} \cdot \pi_j \tag{19}$$

$$\tilde{\alpha}_{ij} = \frac{\exp(\tilde{e}_{ij})}{\sum_{k\in\mathcal{N}(i)} \exp(\tilde{e}_{ik})} \tag{20}$$

This formulation biases the attention mechanism toward nodes with higher structural importance, effectively combining learned attention with global graph priors.

The node embeddings are updated as:

$$h_i^{(l+1)} = \sigma\left(\sum_{j\in\mathcal{N}(i)} \tilde{\alpha}_{ij} W x_j\right) \tag{21}$$

4.6 Slide-Level Prediction

After L layers of graph propagation, we obtain final node embeddings $\{h_i\}_{i=1}^N$. To compute a slide-level representation, we aggregate node predictions using mean pooling:

$$\hat{y} = \sigma\left(\frac{1}{N}\sum_{i=1}^{N} \mathbf{w}^T h_i\right) \tag{22}$$

where $\sigma(\cdot)$ denotes the sigmoid function.

4.7 Importance Map Generation

In addition to classification, we generate interpretable importance maps by combining learned predictions with PageRank scores:

$$s_i = \hat{y}_i \cdot \pi_i \tag{23}$$

where $\hat{y}_i$ is the patch-level prediction. The final importance score s_i reflects both structural centrality and learned relevance.

These scores are projected back to spatial coordinates to produce heatmaps highlighting diagnostically important regions.

4.8 Training Objective

The model is trained using binary cross-entropy loss:

$$\mathcal{L} = -y \log \hat{y} - (1 - y) \log(1 - \hat{y}) \tag{24}$$

where y is the ground-truth slide label.

4.9 Interpretability and Advantages

The proposed framework provides a principled mechanism for combining structural and learned importance. PageRank introduces a global prior that stabilizes attention, while the GNN captures local interactions. This results in more coherent and interpreable importance maps compared to purely data-driven approaches.

5 Experiments and Results

We evaluate ONCO-PRISM in two complementary settings: (i) a patch-level benchmark used to establish graph-learning baselines, and (ii) a slide-level WSI pipeline used to study PageRank-guided interpretability and ablations. Because these settings differ in label granularity and input modality, we report them separately.

5.1 Experimental Setup

Patch-level benchmark. We subsample 5,000 patches from the PatchCamelyon training split and use a fixed train/validation/test partition with a positive-class prevalence of 45.6%. Patches are resized to 128×128, embedded with a pretrained ResNet-18, and connected with an 8-NN graph in feature space. We compare a standard GCN against a PageRank-guided GAT (PRGAT).

WSI pipeline. For the slide-level pipeline, each WSI is tiled into 224×224 patches, background tissue is filtered using an intensity threshold, patch features are extracted with a pretrained ViT-Base backbone, and patch coordinates are connected with a spatial k-NN graph. PageRank is computed on this graph and used either as a standalone score, as a model score, or as a multiplicative fusion of both.

5.2 Evaluation Metrics

We report Accuracy, Balanced Accuracy, Precision, Recall, F1, ROC-AUC, PR-AUC, and MCC. For the WSI pipeline we additionally report threshold sensitivity and calibration behavior, since the operating point can shift depending on whether the objective prioritizes recall or precision. All results use a default decision threshold of 0.5 unless otherwise stated.

5.3 Patch-Level Baselines

Table 1 summarizes the patch-level benchmark. GCN and PRGAT obtain similar overall accuracy, with PRGAT providing a marginal gain in balanced accuracy, recall, and MCC. The two baselines are therefore comparable in aggregate classification performance, but PRGAT better preserves the positive class at the default operating point.

Table 1: Patch-level baseline comparison on the curated PatchCamelyon subset.

Model	Accuracy	Balanced Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	MCC
GCN (Baseline 1)	0.8220	0.8226	0.7908	0.8289	0.8094	0.9061	0.9029	0.6432
PRGAT (Baseline 2)	0.8220	0.8236	0.7837	**0.8421**	**0.8118**	0.9057	0.8956	**0.6449**

5.4 WSI Pipeline Results and Ablation

Table 2 reports the slide-level WSI ablation. The model-only variant is strongest at the default threshold, achieving the highest accuracy, F1, ROC-AUC, PR-AUC, and MCC. In contrast, PageRank alone is not sufficient as a predictor, but it still provides a useful structural prior when combined with the learned model score. The combined PageRank $\times$ Model score trades recall for precision, indicating that the multiplicative fusion yields a more conservative operating point.

Table 2: WSI pipeline ablation on slide-level test data. Best values are bolded.

Variant	Accuracy	Balanced Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	MCC
PageRank only	0.5040	0.4870	0.4351	0.2939	0.3508	0.4791	0.4516	-0.0280
Model only (WSI)	**0.9520**	**0.9523**	0.9397	**0.9561**	**0.9478**	**0.9916**	**0.9906**	**0.9035**
PageRank $\times$ Model (WSI)	0.6880	0.6600	**0.9286**	0.3421	0.5000	0.7671	0.7883	0.4264

The default threshold of 0.5 is used for direct comparison. A threshold sweep on the combined WSI score shows that the best F1 is obtained near 0.2 (F1 $= 0.6843$, accuracy $= 0.7140$), whereas increasing the threshold gradually improves specificity at the expense of recall. This behavior is consistent with the conservative calibration profile observed in the combined score.

5.5 WSI Interpretability

We retain the cancer-importance comparison Figure 1 because it provides the most direct explanation of the proposed spatial prior. The PageRank heatmap identifies structurally central tissue regions, while the model-score overlay highlights discriminative tissue content. Their multiplicative fusion suppresses isolated noise and produces a smoother, more spatially coherent saliency map.

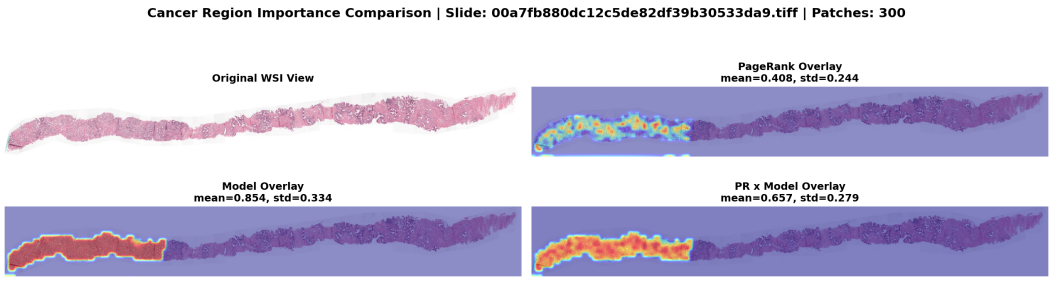

Figure 1: WSI Cancer-importance comparison

5.6 Ablation and Ranking Views

The ablation plot Figure 2 confirms that PageRank alone is not a sufficient classifier, but the model-only branch is strong, and the combined score preserves most of the model's discriminative power while adding structural regularization. The radar plot Figure 2 makes the same conclusion visible across multiple metrics simultaneously.

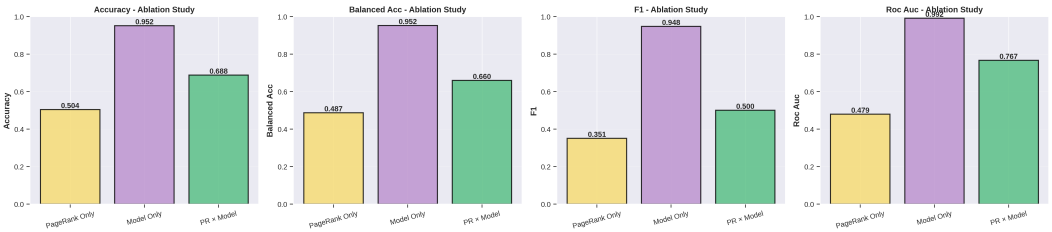

Figure 2: Ablation study showing the contribution of graph structure and PageRank

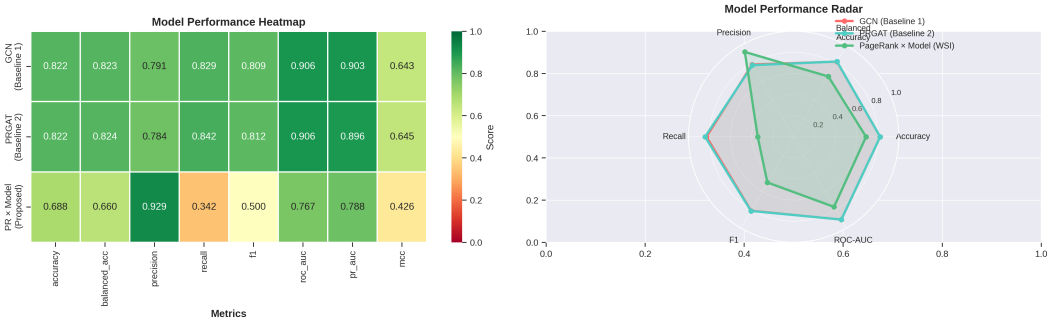

Figure 3: Multi-metric radar comparison illustrating performance across different evaluation metrics.

5.7 Visualizations

The ROC curves Figure 4 show that the patch-level baselines achieve strong separability, while the WSI combined score remains more conservative. The precision-recall curves are more informative under class imbalance and confirm that PRGAT slightly improves positive-class recovery over GCN. The calibration panel reveals that the WSI combined score is better calibrated in the high-confidence regime, while the baseline patch classifiers exhibit sharper score separation.

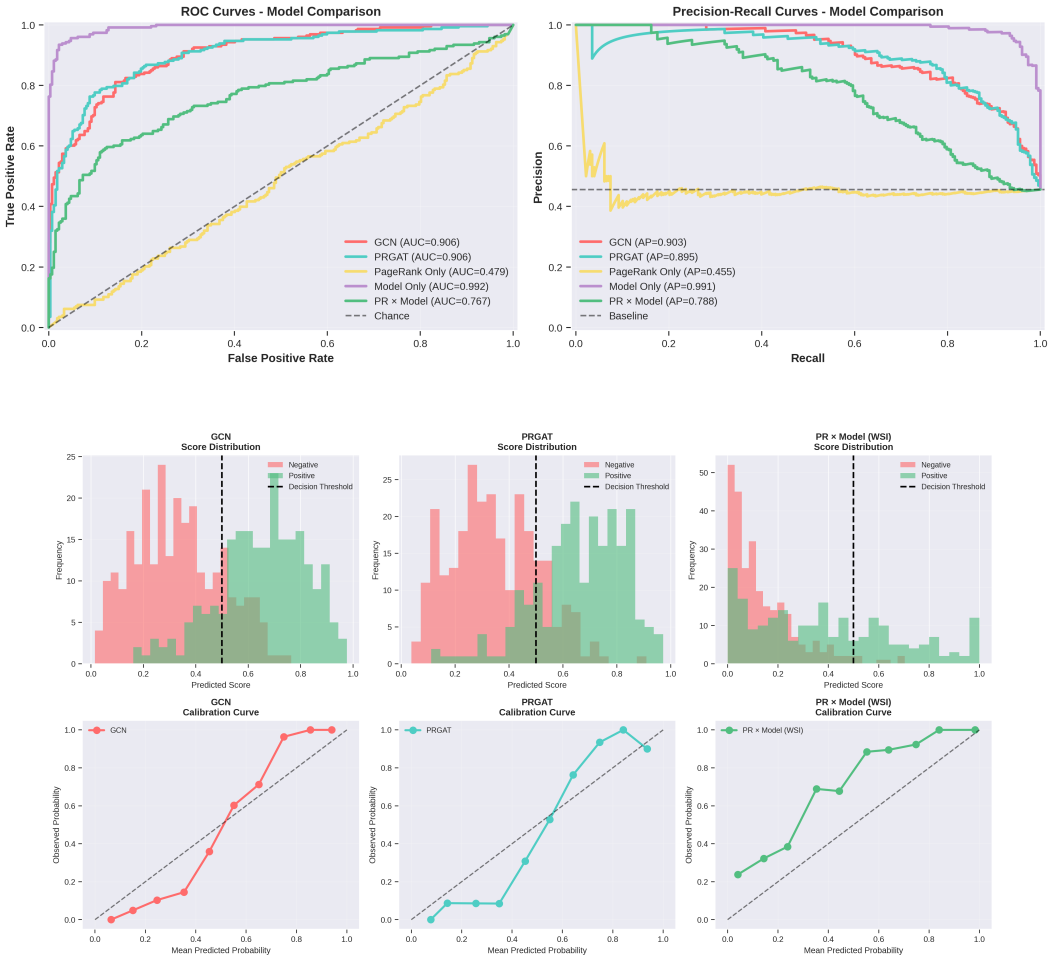

Figure 4: ROC/PR curves and calibration diagnostics for the evaluated models.

Figure 5 complements the aggregate metrics with confusion matrices and error breakdowns. These plots make the operating-point trade-offs explicit: PRGAT slightly increases true positives compared to GCN, while the WSI combined score substantially reduces false positives but misses a larger number of positives at the default threshold.

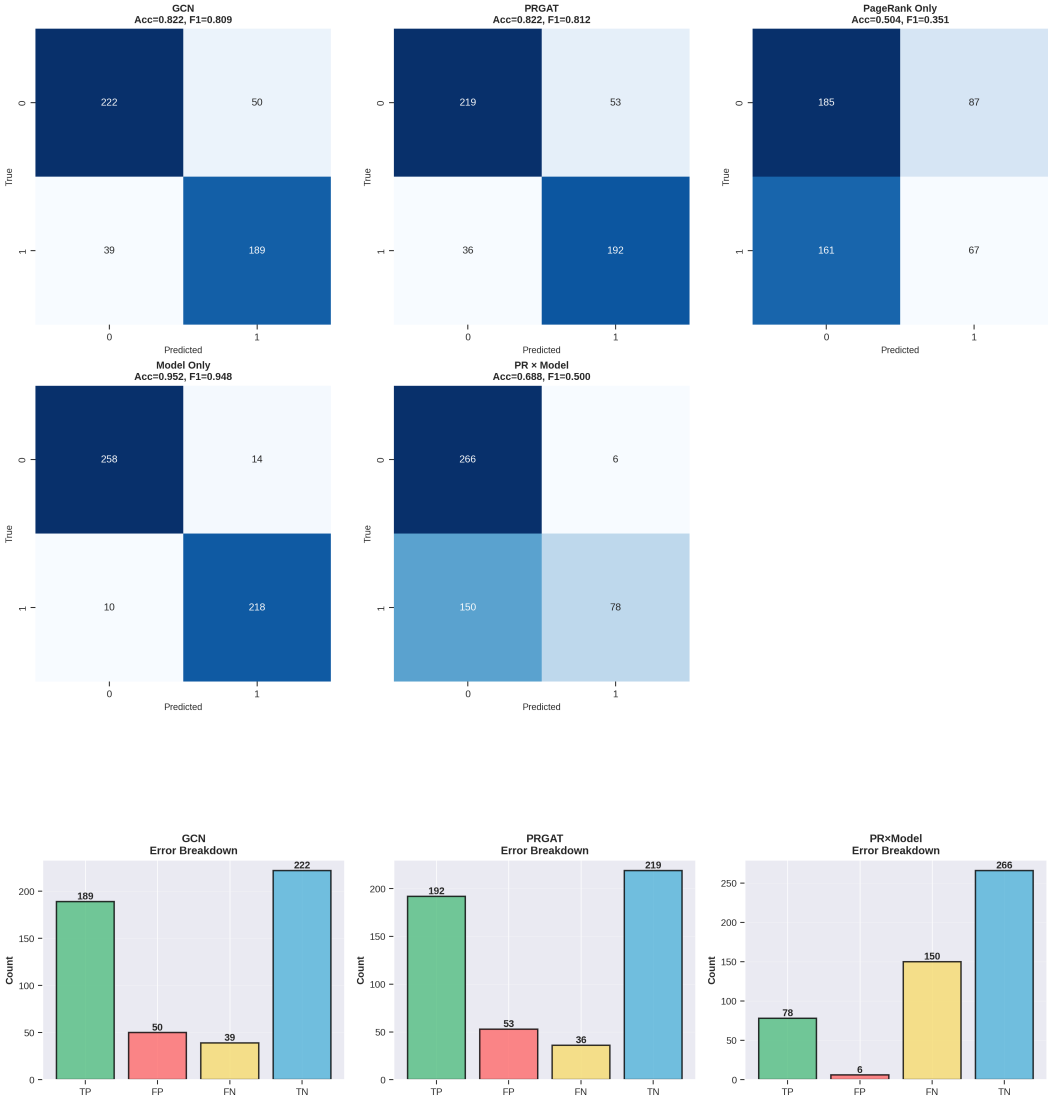

Figure 5: Confusion matrices and error-case decomposition at the default threshold.

6 Conclusion and Future Work

In this work, we introduced ONCO-PRISM, a novel framework for cancer region identification in whole slide images that integrates graph-based modeling with PageRank-guided attention. By representing tissue as a graph of spatially related patches and incorporating PageRank as a measure of global structural importance, our approach effectively combines relational reasoning with principled structural priors.

The proposed method addresses key limitations of existing approaches in computational pathology. While convolutional and transformer-based models process patches largely independently, and existing graph-based methods rely solely on learned attention, ONCO-PRISM explicitly models both local interactions and global importance. This results in improved spatial coherence and more interpretable region importance maps. Through qualitative analysis, we demonstrate that the proposed method produces more localized and meaningful highlights compared to baseline approaches, aligning well with tissue structures observed in histopathology.

Although quantitative improvements are constrained by the limited dataset size used in our experiments, the results consistently indicate that incorporating structural priors enhances both stability and interpretability. The ability of PageRank to guide attention toward structurally significant regions provides a valuable inductive bias, particularly in weakly supervised settings where patch-level annotations are unavailable.

Despite these promising results, several avenues remain for future exploration. First, scaling the proposed framework to larger datasets would enable more robust evaluation and potentially yield stronger quantitative gains. Second, the current formulation assumes a static graph structure; extending this to dynamic or adaptive graph construction could further improve the modeling of complex tissue interactions. Third, integrating multi-scale representations may allow the model to capture both fine-grained cellular details and broader architectural patterns within tissue. Additionally, incorporating domain-specific priors, such as cellular morphology or staining characteristics, could further enhance performance.

Finally, the interpretability offered by the proposed approach opens opportunities for clinical integration. Future work may explore collaboration with pathologists to validate the highlighted regions and assess the model's utility in real-world diagnostic workflows. More broadly, this work demonstrates the potential of combining classical graph-theoretic concepts with modern deep learning architectures, paving the way for more structured and interpretable approaches in medical image analysis.

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